

Component Contributions in ML Agents — A Scaled Reproduction

A T4 proxy tests iteration and fixed-role coordination; a timestamped audit tests K-LIVE’s release claims.



Logbook &
artifacts

Independent reproduction[☒]

Original study: Junsung Kim et al. · OpenReview FOfvTwBGUX

1 What the paper reports

The paper’s K-LIVE ablations report that **iterative feedback** matters more than architectural additions.

+34.1	−8.5	3,973
points from iteration	points from multi-agent	paper trajectories

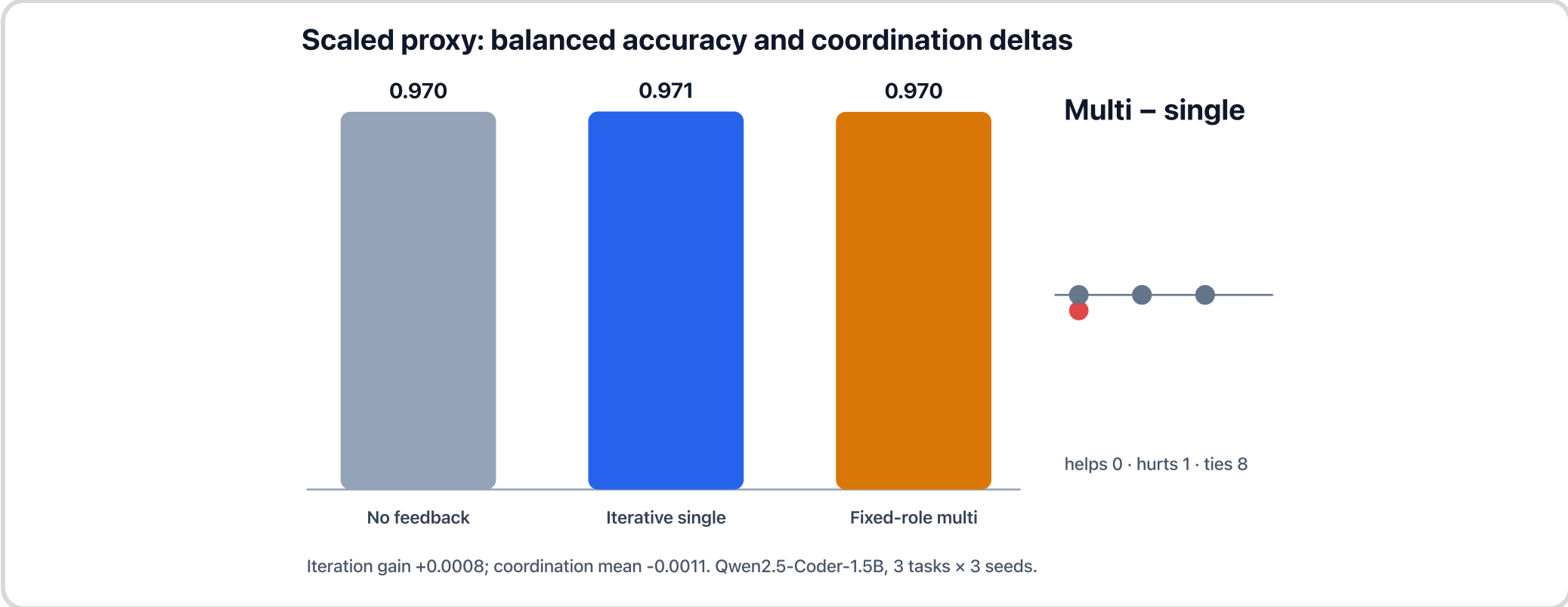
Source confirmation only: these numbers are reported by the paper, not independently reproduced here.

2 Scaled proxy protocol

Backbone	Qwen2.5-Coder-1.5B-Instruct
Tasks	3 local sklearn classification tasks
Seeds	3 per task and condition
Conditions	no feedback; 3-round iteration; fixed-role trio
Metric	balanced accuracy

All 27 measurements ran on one NVIDIA T4. This tests the claimed direction under a transparent, inexpensive proxy—not Kaggle leaderboard performance.

3 GPU proxy results



Mean across 3 tasks × 3 seeds. Error bars show standard deviation across paired deltas. Raw rows: [outputs/gpu/trials.csv](#).

4 Claim 1 verdict ★

+0.078	−0.105
pp iteration (SD 0.155)	pp coordination (SD 0.316)

INCONCLUSIVE / PROXY DOES NOT SUPPORT. Iteration did not dominate coordination in magnitude. The proxy is too small and structurally different to falsify the full study.

5 Can coordination hurt?

Fixed-role Explorer–Builder–Evaluator is compared with iterative single-agent feedback for each task–seed pair.

0	1	8
pairs helped	pair hurt	ties

PARTIAL / PROXY SUPPORT. Coordination can hurt, but nine mostly tied pairs cannot establish “as often as it helps.”

6 K-LIVE inventory audit

On 2026-07-18, all 25 paper-listed Kaggle URLs resolved: **13 Tier-1 deadline competitions** + **12 Tier-2 perpetual competitions**.

Model cutoff	Pre	Post	Active
Gemini 2.0 Pro	13	0	0
DeepSeek V3	13	0	0
Claude 3.7	13	0	0
Kimi K2	0	8	5

The counts exactly match the paper’s cutoff table.

7 “Contamination-free” is too strong

- All 13 Tier-1 deadlines had passed.
- The cited public K-LIVE repository returned **404 / repository not found**.
- Date cutoffs cannot exclude indirect exposure.
- The paper itself calls its guarantees “not airtight.”

PARTIAL / CLAIM OVERSTATED. The 25-item inventory and cutoff matrix verify; current dynamism and contamination freedom do not.

8 Scope, cost & takeaways ★

T4	12m20s	≈\$0.10
16 GB GPU	successful Job	all T4 attempts

- **Claim 1:** inconclusive under the scaled proxy.
- **Claim 2:** coordination hurt one pair (weak support).
- **Claim 3:** count confirmed; strongest wording not established.

Reproducible boundary: scripts, 27 raw trials, immutable Job logs, audit outputs, poster, and source notes are bundled in the public logbook.